

Overview

This course builds on DS110 (Python for Data Science) by expanding on programming language, systems, and algorithmic concepts introduced in the prior course. The course begins by introducing shell commands, using command windows and git version control. These are practical skills that are essential a practicing data scientist.

You will then explore the different types of programming languages and be introduced to important systems level concepts such as computer architecture, compilers and file systems. It is vital to conceptualize how programs work at the machine level.

The bulk of the course is spent learning Rust, a modern, high-performance and more secure programming language. Rust is a systems programming language that is designed to be safe, fast, and memory efficient. It is a great language to learn because it is a low-level language that is still easy to read and write. More and more performant data science libraries and tools are written in Rust for these reasons.

You will be expected to read relevant parts of the Rust Language Book before each lecture, where we will then present the material in more depth. You will then have the opportunity to practice what you just learned with in-class activities. There will be approximately seven homeworks, two midterms, and a final exam.

Learning any new programming language is significant time and effort investment and it is vital to continually practice what you learn throughout the entire semester.

Prerequisites: CDS 110 or equivalent

B1 Course Staff

Section B1 Instructor: Thomas Gardos

Email: tgardos@bu.edu

Office hours: 2-3pm Tuesdays and Thursdays @ CCDS 1623, and by appointment.

If you want to meet but cannot make office hours, send a private note on Piazza with at least 2 suggestions for times that you are available, and we will find a time to meet.

B1 TAs

See Piazza resource page for office hours and contact information.

- Gabriel Maayan
- Zachary Gentile

B1 CAs

See Piazza resource page for office hours and contact information.

- Emir Tali
- Matthew Morris
- Kesar Narayan
- Lingjie Su

Lectures and Discussions

B1 Lecture: Tuesdays, Thursdays 11:00am-12:15pm (SHA 110)

Section B Discussions (Fridays, 50 min):

- B2: Fri 12:20pm – 1:10pm, IEC B10 (888 Commonwealth Ave.)
 - B3: Tue 1:25pm – 2:15pm, CGS 313 (871 Commonwealth Ave.)
 - B4: Tue 2:30pm – 3:20pm, CDS 164 (665 Commonwealth Ave.)
 - B5: Tue 3:35pm – 4:25pm, CDS 164 (665 Commonwealth Ave.)
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Note: There are two sections of this course, they cover similar material but the discussion sections and grading portals are different. These are not interchangeable, you must attend the lecture and discussion sessions for your section!

Course Websites

Links shared via email.

- Piazza
 - Lecture Recordings
 - Announcements and additional information
 - Questions and discussions
- Course Notes:
 - Syllabus (this document)
 - Interactive lecture notes
- Gradescope
 - Homework, project, project proposal submissions
 - Gradebook
- GitHub Classroom: URL TBD

Course Content Overview

For a complete list of modules and topics that will be kept up-to-date as we go through the term, see [B1 Lecture Schedule \(TTH\)](#).

Course Format

Lectures will involve extensive hands-on practice. Each class includes:

- Interactive presentations of new concepts
- Small-group exercises and problem-solving activities
- Discussion and Q&A

Because of this active format, regular attendance and participation is important and counts for a significant portion of your grade (15%).

Discussions will review lecture material, provide homework support, and will adapt over the semester to the needs of the class. We will not take attendance but our TAs make this a great resource!

Pre-work will be assigned before most lectures to prepare you for in-class activities. These typically include readings plus a short ungraded quiz. We will also periodically ask for feedback and reflections on the course between lectures.

Homeworks will be assigned roughly weekly at first, and there will be longer two-week assignments later, reflecting the growing complexity of the material.

Exams Two midterms and a cumulative final exam covering theory and short hand-coding problems (which we will practice in class!)

The course emphasizes learning through practice, with opportunities for corrections and growth after receiving feedback on assignments and exams.

Course Policies

Grading Calculations

Your grade will be determined as:

- 15% homeworks (~9 assignments)
- 20% midterm 1
- 20% midterm 2
- 25% final exam
- 15% in-class activities and attendance polls
- 5% pre-work and surveys

We will use the standard map from numeric grades to letter grades (≥ 93 is A, ≥ 90 is A-, etc).

For the midterm and final, we may add a fixed number of "free" points to everyone uniformly to effectively curve the exam at our discretion - this will never result in a lower grade for anyone.

We will use gradescope to track grades over the course of the semester, which you can verify at any time and use to compute your current grade in the course for yourself.

Homeworks

Homework assignments will be submitted by uploading them to GitHub Classroom. We will use Rust tests and GitHub Actions to automatically test your code. We'll also inspect for evidence of good git version control practices. You will get more instructions on homeworks in class and on Piazza.

You are expected to complete homeworks yourself and not have AI do it for you. Per the AI policy below, you are allowed to use AI to help you understand concepts, debug your code, or generate ideas. You should understand that this may help or impede your learning depending on how you use it.

If you use AI for an assignment, you must cite what you used and how you used it (for brainstorming, autocomplete, generating comments, fixing specific bugs, etc.). You must understand the solution well enough to explain it during a small group or discussion in class. You should be able to explain your code to a peer in a way that is easy to understand.

Your professor and TAs/CAs are happy to help you write and debug your own code during office hours, but we will not help you understand or debug code that is generated by AI.

For more information see the [CDS policy on GenAI](#).

Exams

The final will be during exam week, date and location TBD. The two midterms will be in class

during normal lecture time.

If you have a valid conflict with a test date, you must tell me as soon as you are aware, and with a minimum of one week notice (unless there are extenuating circumstances) so we can arrange a make-up test.

If you need accommodations for exams, schedule them with the Testing Center as soon as exam dates are firm. See below for more about accommodations.

Deadlines and late work

Homeworks will be due on the date specified in gradescope and github classroom.

If your work is up to 48-hours late, you can still qualify for up to 80% credit for the assignment. After 48 hours, late work will not be accepted unless you have made prior arrangements due to *extraordinary* circumstances.

Because of our autograding system, it is possible to get partial credit for homework submitted on time, and then 80% credit for remaining work submitted up to 48 hours late.

Collaboration

You are free to discuss problems and approaches with other students but must do your own writeup. If a significant portion of your solution is derived from someone else's work (your classmate, a website, a book, etc), you must cite that source in your writeup. You will not be penalized for using outside sources as long as you cite them appropriately.

You must also understand your solution well enough to be able to explain it if asked.

Academic honesty

You must adhere to BU's Academic Conduct Code at all times. Please be sure to read it [here](#). In particular: cheating on an exam, passing off another student's work as your own, or

plagiarism of writing or code are grounds for a grade reduction in the course and referral to BU's Academic Conduct Committee. If you have any questions about the policy, please send me a private Piazza note immediately, *before* taking an action that might be a violation.

AI use policy

You are allowed to use GenAI (e.g., ChatGPT, GitHub Copilot, etc) to help you understand concepts, debug your code, or generate ideas. You should understand that this may help or impede your learning depending on how you use it.

If you use GenAI for an assignment, you must cite what you used and how you used it (for brainstorming, autocomplete, generating comments, fixing specific bugs, etc.). You must understand the solution well enough to explain it during a small group or discussion in class.

Your professor and TAs/CAs are happy to help you write and debug your own code during office hours, but we will not help you understand or debug code that generated by AI.

For more information see the [CDS policy on GenAI](#).

Attendance and participation

Since a large component of your learning will come from in-class activities and discussions, attendance and participation are essential and account for 15% of your grade.

Attendance will be taken in lecture through Piazza polls which will open at various points during the lecture. Understanding that illness and conflicts arise, up to 4 absences are considered excused and will not affect your attendance grade.

In most lectures, there will be time for small-group exercises. To receive participation credit on these occasions, you must submit a group assignment on Gradescope. These submissions will not be graded for accuracy, just for good-faith effort.

Occasionally, I may ask for volunteers, or I may call randomly upon students or groups to answer questions or present problems during class.

Absences

This course follows BU's policy on religious observance. Otherwise, it is generally expected that students attend lectures and discussion sections. If you cannot attend classes for a while, please let me know as soon as possible. If you miss a lecture, please review the lecture notes and lecture recording. If I cannot teach in person, I will send a Piazza announcement with instructions.

Accommodations

If you need accommodations, let me know as soon as possible. You have the right to have your needs met, and the sooner you let me know, the sooner I can make arrangements to support you.

This course follows all BU policies regarding accommodations for students with documented disabilities. If you are a student with a disability or believe you might have a disability that requires accommodations, please contact the Office for Disability Services (ODS) at (617) 353-3658 or access@bu.edu to coordinate accommodation requests.

If you require accommodations for exams, please schedule that at the [BU testing center](#) as soon as the exam date is set.

Re-grading

You have the right to request a re-grade of any homework or test. All regrade requests must be submitted using the Gradescope interface. If you request a re-grade for a portion of an assignment, then we may review the entire assignment, not just the part in question. This may potentially result in a lower grade.

Corrections

You are welcome to submit corrections on midterms. This is an opportunity to take the feedback you have received, reflect on it, and then demonstrate growth.

We will provide solutions as part of the midterm grading process, so simply resubmitting the solution will earn you no credit.

Instead, what we are looking for is a personal reflection written in your own words that addresses the following:

- A clear explanation of the mistake
- What misconception(s) led to it
- An explanation of the correction
- What you now understand that you didn't before

After receiving grades back, you will have one week to submit corrections. You can only submit corrections on a good faith attempt at the initial submission (not to make up for a missed assignment).

Satisfying this criteria completely for any particular problem will earn you back 50% of the points you originally lost (no partial credit).

The Rust Language Book

The primary reference will be the [Rust Language Book](https://rust4ds.github.io/ds210-sp26-b1-lectures/index.html) and these course notes.